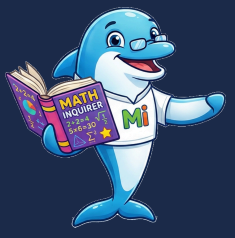
TUTOR GUIDE • LEVEL 1 • TERM 4 • LESSON 2 OF 10

Build the Figure



Big idea: Make big shapes from small shapes, and find the small shapes inside a big one. Put shapes together to make a figure. Look inside a figure to find the shapes it is made of.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	2D Shapes: forming figures and finding the shapes in a figure (Primary 1) (Form figures from given shapes, and identify the shapes in a figure (Primary 1, 2D Shapes 1.2 and 1.3, p. 32); Count a named shape in a picture made of shapes (seen in Primary 1 school papers); Counting every triangle of every size in a figure with crossing lines, and the fewest pieces that make a figure, are not Primary 1 outcomes) Strand label for the tutor only. Start here is Primary 1 shapes work: count each kind of shape in a picture, cut a square into 2 triangles, build a boat from squares and triangles, read a rectangle as rows of squares, and say what 2 shapes make together (2 squares, 2 half circles, 2 quarter circles). Think harder splits a figure into its shapes with a ruler (DIAG), counts every shape of each kind into a table (TAB), and meets the beyond puzzle Every Triangle: a square with both corner-to-corner lines holds 8 triangles, 4 small and 4 big. Go further asks for the fewest pieces (a 2 by 3 rectangle needs 3, not 6), a list of the piece sets that make a figure with a check that none is missing (LIST), and a figure to make for a partner. BEYOND PRIMARY 1 ON PURPOSE, at Think harder and Go further only: counting hidden and overlapping triangles, the fewest pieces, and which sets of pieces make a figure. No Primary 1 school paper in the sample we studied asks a child to build a figure from pieces (the papers count and name shapes in a picture), so the building wording is marked unverified. Do not call this lesson MOE-aligned. Owner decision, 22 and 23 Oct 2026: strong content with scaffolding, said plainly to families; every lesson names a beyond item and a non-routine type. Contest tag (tutor only): counting figures (SASMO, 6 of 100).
Objectives	• Count each kind of shape in a picture and in a figure, and say how many more of one kind than another. • Put 2 triangles together, whole side to whole side, to make a square or a bigger triangle, and say what 2 squares, 2 half circles or 2 quarter circles make. • Build a figure from squares and triangles and say how many pieces of each kind it needs. • Think harder: split a figure into its shapes with a ruler, count shapes of each kind into a table, and count every triangle in a square with both corner lines. • Go further: find the fewest pieces that make a figure, say which sets of pieces make it, and draw a figure for a partner to count.
Joining cold	**Remember warm-up covers:** Name a triangle, a square and a rectangle by their sides and corners (the holiday course, taught again with clues); Fold a paper square corner to corner and name the 2 shapes (the first lesson of Term 4, taught again with a spare square); Count the squares in a rectangle by rows (Term 2 Lesson 5, taught again with a picture). Only counting to 12 and the names of three shapes. Nothing from an earlier lesson is needed: the warm-up names shapes from clues, folds a square corner to corner and counts squares by rows. The words whole side, piece and figure are taught in Explore and the Mini-lesson. Fairly safe to miss: no later lesson needs it except Count Everything (Term 4 Lesson 6), which counts squares in a grid and has its own warm-up, and the year Quest, which has a gate on shapes. A child who misses it watches the recording, puts 2 triangles together every way they can and does workbook pages 1 and 2. A child who joins here cold can follow: the warm-up re-teaches the shape names, the corner fold and counting by rows.

Child materials	• Your Term 4 book (Book 7): the 2 shape-pieces pages (pages 6 and 7), torn out along the dashed line, with the 20 pieces cut out by a grown-up beforehand: 6 squares, 3 rectangles, 6 triangles, 2 half circles, 2 quarter circles and 1 circle. Nobody cuts during the lesson • No printer or no cutting? A grown-up cuts 6 squares (3 cm each side), 3 rectangles (3 cm by 6 cm) and 6 triangles (cut a 3 cm square corner to corner) from card or paper, and any round shapes you like • A paper square from last lesson (or any paper square), a ruler (the one you keep at home), a pencil, a rubber and spare paper • A coloured pencil is helpful
Tutor materials	• The slide deck • Your own set of pieces to build a figure on camera, if you wish • A visible clock A ruler is used to draw straight lines (Start here S2 and Think harder T1). Nothing needs to be bought. The pieces are small and have corners: they stay on the table, nothing goes in the mouth or ears, and they are kept away from younger brothers and sisters. No scissors, matches, toothpicks or anything sharp in the lesson: a grown-up cuts the pieces beforehand. In class we use the squares, rectangles and triangles to build figures; the half circles, quarter circles and the circle are for Start here S5 and for free building. All the straight pieces are on the same 3 cm unit (a rectangle is 2 squares, a triangle is half a square).
Tiers	Start here (5 questions, pages 1 and 2), Think harder (3, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit pages (pages 6-7). Labels are neutral; nobody is told which page they are on.
Beyond Primary 1 (tutor only)	Every Triangle at T3, G1, G2: A square with both corner-to-corner lines holds 8 triangles: 4 small ones next to the middle and 4 big ones, each made of 2 small ones (T3). Counting triangles of every size is not a Primary 1 outcome; naming and counting shapes in a picture is. G1 and G2 ask for the fewest pieces (a 2 by 3 rectangle needs 3, not 6; the staircase needs 4) and which sets of pieces make a figure (the LIST type): enrichment at Go further only. T1 (split a figure with a ruler) is the DIAG item and T2 (count each kind into a table) the TAB item. Named puzzle: Every Triangle. Never in Start here, never advertised to children.
Contest-style move	Used at Think harder, Go further and the Investigation: table: count each kind of shape in a figure into a table (TAB); draw it: split a figure into its shapes with a ruler (DIAG); count every triangle of every size in a figure (Every Triangle); list: which sets of pieces make this figure, with a check that none is missing (LIST); fewest pieces: the same figure made with the smallest number of pieces. Never advertised to children or parents.

Before the lesson

- Solve every workbook question yourself first, with your own pieces and a ruler. Check: S1 4, 3, 1 and 1 more; S2 a corner-to-corner line and half; S3 2 triangles and 2 squares; S4 4, 3, 12; S5 rectangle, circle, half circle; T1 one line across the bottom of the roof, then 2 more lines; T2 triangles 5, squares 6, circles 3; T3 8 (4 small and 4 big); G1 6 and 3; G2 true, false, true, false; G3 any figure with exactly 2 triangles, 1 square and 1 rectangle; I1 a square first, then 3 different shapes in all; E1 7; E2 Yes; E3 3.
- Tell families before this lesson (the goodbye of the first lesson of Term 4 says so): the shape-pieces pages (pages 6 and 7 of the Term 4 book) are torn out along the dashed line; a grown-up cuts out the 20 pieces (6 squares, 3 rectangles, 6 triangles, 2 half circles, 2 quarter circles, 1 circle) and keeps them in an envelope; a paper square from last lesson and a ruler. Print the kit at 100 percent so a square keeps its 3 cm sides. No printer: a grown-up cuts 6 squares (3 cm), 3 rectangles (3 cm by 6 cm) and 6 triangles (a 3 cm square cut corner to corner) from card or paper.
- Have the slides open. Have your own pieces to build on camera if you wish. Say piece, whole side, kind of shape and in all the same way every time. Do not say that 2 triangles always make a square. Do not write fraction notation.
- Check the recording captures the tutor and the shared screen. Children's video is excluded; any unmuted voice is recorded. Greet each child by first name as they arrive. Say the recording notice, then start the recording; once it is running, avoid first names. Nobody holds paper or a figure up to the camera.
- Be plain with families: counting and naming shapes in a picture, and putting shapes together, are school Primary 1 work (2D Shapes 1.2 and 1.3). No Primary 1 school paper in the sample we studied asks a child to build a figure from pieces, so the building wording is marked unverified. Counting every triangle of every size (Every Triangle), the fewest pieces and which sets of pieces make a figure go beyond Primary 1 on purpose, at Think harder and Go further only. Do not call the lesson MOE-aligned.
- Slides s-explore1, s-explore2, s-m2, s-m3 and s-m5 keep the answer off the screen while you ask: it appears on the slide after. Read every question aloud, including the numbers under the shapes and the shape names: the pictures on the slides are small.

Safeguarding and recording (this lesson)

- The shape pieces are small and have corners: they stay on the table, nothing goes in the mouth or ears, and they are kept away from younger brothers and sisters. No scissors, matches, toothpicks or anything sharp. Nobody cuts during the lesson; a grown-up cuts the 20 pieces beforehand.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Nobody holds paper or a figure up to the camera: a gallery view shows everyone. Children say what they built, tick their own work against the answer slide, answer together on a count of three, or show fingers or a thumb for a choice. There is no private chat.
- Never read out a child's name with a right or wrong answer. The recording has the tutor, the shared screen and any voice that is unmuted. Children may mute and keep their cameras off. A child's page is never shown.
- Movement break: children need a little clear space, soft shoes or socks on a safe floor, no stairs and no running. Making shapes with the arms is quiet and small; a child may stay seated and make the shapes with their hands.
- A child who is still stuck ticks 'I am still stuck' on the look-back page. A grown-up may photograph that page after class and send it to the business WhatsApp only; photos are kept for 90 days after the program ends. The tutor then helps in the open session next lesson. No private messages to children, and no personal phones.
- The look-back is never collected in front of the group or read out. A pass is always allowed.
- The gallery shows faces and fingers, not tables or papers. Ask for a finger answer when you need to see something. Answers on three are said quietly with the microphone off; a child who is invited to share switches the microphone on, then off again. The still-stuck photo is of the whole of workbook page 5 (Investigation, exit ticket and look-back).

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills

0–2	**Opening**	2	1: Build the Figure	1:41
2–8	**Remember warm-up**	6	3: Which shape am I?, Fold a square corner to corner, Count by rows	4:58
8–12	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	2:57
12–24	**Explore**	12	5: Explore: 2 triangles, Explore: 2 triangles, Explore: build the house, Explore: build the house, Explore: your own picture	8:38
24–36	**Mini-lesson**	12	8: 2 shapes make 1 shape, Find the shapes inside, Find the shapes inside, Count each kind, Count each kind, A rectangle can replace 2 squares, Your turn, Your turn	9:24
36–40	**Brain break**	4	1: Shape moves	3:10
40–70	**Workbook**	30	5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page	21:12
70–76	**Talk time**	6	2: Talk and explain, Back to the sneak-peek	4:22
76–82	**Exit ticket**	6	2: Exit ticket, Check the exit ticket	4:34
82–85	**Look-back**	3	1: Look-back	2:23
85–90	**Flex time and goodbye**	5	1: Well done, inquirers!	0:30
	Total	**90**		**63:48**

The script needs 63:48 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 75:30. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 12 (end of sneak-peek), minute 40 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 40, skip in this order: Start with: shorten Explore round 3 (your own picture) to 3 minutes and carry the counting by kind to the Mini-lesson; Then: cut the second item of the Your turn slide (s-m5); Then: take the extra minutes from the flex time. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 2:00 Opening

script 1:41 of 2:00

0:00	SLIDE	Show slide **Build the Figure** (s-title)
0:08	DO	Greet each child by first name as they arrive, before the recording starts. Check that you can see each child. [0:15]
0:23	SAY	Hello, inquirers! Pencil ready, and your 20 shape pieces on the table. Cameras can be on or off, as you like.
0:34	SAY	This lesson is recorded so a friend who is away can catch up. The recording has me, the shared screen and any voice that is not muted. You may keep your microphone and camera off.
0:51	DO	Start the recording now. Do not use children's first names on the recording. [0:05]
0:56	REC	*If you are watching the recording: hello! Get a pencil, a ruler, paper and your 20 shape pieces. Pause or rewind any time.*
1:08	SAY	Microphone off until I ask you to speak. To answer, say it with me on three, show fingers, or give a thumbs up. If I invite you to talk to everyone, switch your microphone on, then off again. Nobody holds paper up to the camera.
1:30	SAY	Today we put small shapes together to make big shapes, and we look inside a big shape to find the small ones.
	NOTE	Say the recording notice before you start the recording, and do not use children's first names once it is running. Say the recording statement and the microphone routine in the same words every lesson, for anyone who joined late. The routine and the four inquirer rules were introduced earlier in the term: do not re-teach them.
	NOTE	Plain statement for families when it comes up: counting and naming shapes in a picture, and putting shapes together, are school Primary 1 work. Counting every triangle of every size, finding the fewest pieces and listing which sets of pieces make a figure go beyond Primary 1 on purpose, at Think harder and Go further only. The pieces are never cut in the lesson: a grown-up cuts them beforehand.
	NOTE	New here? Say in one minute: our three rule words (notice it, test it, explain why); a guess is fine and you can always pass; after Start here you may choose Think harder, Go further or the Investigation; nobody holds paper or a figure to the camera. A child without the pieces folds paper squares and draws shapes.

2:00 – 8:00 Remember warm-up

script 4:58 of 6:00

2:00	SLIDE	Show slide **Which shape am I?** (s-rem1)
2:25	REC	*Recording: join in! Name each shape, fold a paper square corner to corner, and count the squares by rows.*
2:34	SAY	Three clues, three shapes. Look at shapes A, B and C.
2:40	ASK	Which shape has 3 corners? [wait about 12 s for answers]
2:54	ASK	Which shape has 4 sides that are all the same? [wait about 12 s for answers]
3:12	ASK	Which shape has 2 long sides and 2 short sides? [wait about 14 s for answers]
3:30	ASK	What are the three shapes called? [wait about 20 s for answers]
3:54	SLIDE	Show slide **Fold a square corner to corner** (s-rem2)
4:18	SAY	Take a paper square and fold it corner to corner. Open it and trace the fold.

4:26	**CHILDREN**	Children fold a paper square corner to corner, open it and trace the fold. [0:45]
5:12	**ASK**	What 2 shapes do you see? How do you know they match? [wait about 25 s for answers]
5:42	**SLIDE**	Show slide **Count by rows** (s-rem3)
6:08	**SAY**	Now count the squares in the rectangle, row by row.
6:12	**ASK**	How many squares are in each row? How many rows? [wait about 18 s for answers]
6:36	**ASK**	How many squares are there in all? [wait about 12 s for answers]
6:51	**SAY**	Shapes can be put together and taken apart. That is our idea for today.
	IF STUCK	If a child is unsure of a shape name, point to the shape and say its name. For corners, touch each corner and count. If a child cannot fold corner to corner, fold the paper once so two edges meet and trace the fold. For counting, point to each square and count in ones.
	NOTE	Answers: A the triangle, B the square, C the rectangle; the fold makes 2 triangles that match; 3 squares in each row, 2 rows, 6 squares in all. Children answer with a whisper, fingers or hands. Nobody writes. Joining cold: all you need is counting to 6 and the shape names.

8:00 – 12:00 Sneak-peek puzzles

script 2:57 of 4:00

8:00	**SLIDE**	Show slide **Sneak-peek puzzles** (s-sneak)
8:25	**REC**	*Recording: pause now and try both puzzles. Use 2 triangles for the second one. Guessing is fine. Press play for the next part.*
8:36	**SAY**	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked.
8:49	**READ**	Puzzle one. Look at the picture. How many triangles are there? Write the number.
8:57	**CHILDREN**	Children count the triangles and write the number on their own paper. [0:30]
9:27	**READ**	Puzzle two. Put 2 triangles from your kit together so that whole sides touch. Name one shape you can make. Draw it.
9:39	**CHILDREN**	Children try 2 triangles on the table and draw or write one shape. [1:10]
10:49	**SAY**	Keep your paper. Nobody shows it. We will check our own thinking later, at Talk time.
	NOTE	Puzzle 1 answer: 2. Puzzle 2 answer: many answers are right (a square, a bigger triangle or a slanted four-sided shape). Look for good ideas to praise later (a child who said it depends on which sides touch). Do not write scores or keep a list.

12:00 – 24:00 Explore

script 8:38 of 12:00

12:00	**SLIDE**	Show slide **Explore: 2 triangles** (s-explore1)
12:08	**REC**	*Recording: pause now. Put 2 triangles together, build the house from the picture, then make your own picture.*
12:17	**SAY**	Today your pieces are the teacher. Take 2 triangles from your kit. Put them together so that whole sides touch. Make a square.
12:28	**CHILDREN**	Children put 2 triangles together to make a square. [0:50]
13:18	**ASK**	Which sides touched to make the square? [wait about 18 s for answers]
13:40	**SAY**	Now make a bigger triangle with the same 2 triangles.
13:45	**CHILDREN**	Children make a bigger triangle from the 2 triangles. [0:45]
14:30	**ASK**	What shapes did you make? Did anyone find a third way? [wait about 22 s for answers]
14:58	**SLIDE**	Show slide **Explore: 2 triangles** (s-explore1-ans)
15:06	**SAY**	2 triangles make a square. 2 triangles make a bigger triangle. Whole sides must touch.
15:13	**SLIDE**	Show slide **Explore: build the house** (s-explore2)

15:21	**SAY**	Round two. Here is a house on dots. Build it with your squares and triangles. Then tell me which pieces you used.
15:32	**CHILDREN**	Children build the house on the table from the picture. [1:25]
16:57	**ASK**	Which pieces did you use? How many of each? [wait about 25 s for answers]
17:26	**SLIDE**	Show slide **Explore: build the house** (s-explore2-ans)
17:34	**SAY**	The house uses 2 squares and 2 triangles. That is 4 pieces. The slanted sides of the roof need triangles.
17:44	**SLIDE**	Show slide **Explore: your own picture** (s-explore3)
17:52	**SAY**	Round three. Make a picture of your own: a robot, a rocket or a boat. Use at least 3 different kinds of shape. Then count how many of each shape you used.
18:08	**CHILDREN**	Children make their own picture with up to 8 pieces and count each kind. [1:40]
19:48	**ASK**	Who would like to say: my picture has this many squares, this many triangles, and so on? Volunteers first. [wait about 40 s for answers]
	IF STUCK	A child who cannot start builds a robot: a rectangle for the body, a square for the head and 2 triangles for the feet. A child whose pieces do not touch whole side to whole side is asked to slide them until the sides match. A child with no kit pieces folds paper squares corner to corner to make triangles.
	NOTE	Main room only; children say what they have made. Tilting the camera to the table is allowed; a face need not be in view; nothing is held up to the camera. Answers: a square; a bigger triangle; the house is 2 squares and 2 triangles (4 pieces); a child with a rectangle for the bottom of the house has used 3 pieces, which is fine. Do not say the answer before the children do. Do not name a child with a wrong answer.

24:00 – 36:00 Mini-lesson

script 9:24 of 12:00

24:00	**SLIDE**	Show slide **2 shapes make 1 shape** (s-m1)
24:08	**REC**	*Recording: keep watching and copy the examples. Rewind if you miss a step.*
24:14	**SAY**	Two shapes can make one shape. Put 2 squares side by side. Whole sides touch.
24:22	**ASK**	What shape do 2 squares make? [wait about 15 s for answers]
24:40	**SAY**	Put the 2 half circles together, with their straight sides touching.
24:46	**ASK**	What shape do 2 half circles make? [wait about 15 s for answers]
25:04	**SAY**	And 2 triangles can make a square or a bigger triangle. It depends which sides touch.
25:12	**CHILDREN**	Children try 2 squares, 2 half circles and 2 triangles with their pieces. [1:00]
26:12	**SLIDE**	Show slide **Find the shapes inside** (s-m2)
26:20	**SAY**	Now we look inside a figure. This house is made of shapes. Count each kind.
26:28	**ASK**	How many triangles are in the house? [wait about 15 s for answers]
26:46	**ASK**	How many rectangles? Remember, the wall and the door. [wait about 15 s for answers]
27:06	**ASK**	How many squares? [wait about 12 s for answers]
27:19	**ASK**	How do you know the window is a square? [wait about 18 s for answers]
27:42	**SLIDE**	Show slide **Find the shapes inside** (s-m2-ans)
27:50	**SAY**	1 triangle, 2 rectangles and 1 square. The window is a square because it is as tall as it is wide.
28:00	**SLIDE**	Show slide **Count each kind** (s-m3)
28:08	**SAY**	Count each kind, one kind at a time. Touch each shape as you count it, so you never count one twice.
28:18	**ASK**	How many triangles are there? [wait about 15 s for answers]
28:36	**ASK**	How many circles? How many squares? [wait about 15 s for answers]
28:54	**CHILDREN**	Children touch each shape once on their screen and count each kind quietly. [0:25]

29:19	**ASK**	How many shapes are there in all? Which kind is there most of? [wait about 20 s for answers]
29:46	**SLIDE**	Show slide **Count each kind** (s-m3-ans)
29:54	**SAY**	4 triangles, 1 circle and 1 square. That is 6 shapes in all.
30:00	**SLIDE**	Show slide **A rectangle can replace 2 squares** (s-m4)
30:08	**SAY**	The same figure can be made with different pieces. Here are 4 squares in a row. A rectangle in our kit is as long as 2 squares.
30:22	**CHILDREN**	Children make a row of 4 squares, then the same row with 2 rectangles. [0:50]
31:12	**ASK**	How many pieces with squares? How many with rectangles? Which uses fewer? [wait about 25 s for answers]
31:42	**SLIDE**	Show slide **Your turn** (s-m5)
31:50	**SAY**	Your turn. Item one: a picture has 3 circles and 2 squares. How many shapes are there? Item two: can 2 squares make a triangle? Say it on three, or write it on your paper.
32:08	**CHILDREN**	Children write or say the two answers. Wait 10 seconds before helping. [1:00]
33:08	**SLIDE**	Show slide **Your turn** (s-m5-ans)
33:16	**READ**	Item one: 5 shapes. Item two: No. Tick your own work with a tick.
	NOTE	Key words to use every time: piece, whole side, figure, kind of shape, in all. Say it the same way each time: the figure has 2 squares and 2 triangles, so 4 pieces. Do not use the word wrong for a child, only for a picture or a statement.
	IF STUCK	If a child cannot find the triangles in the house, ask them to trace the pointed shape on top. If a child counts a shape twice, ask them to touch each shape once and move on. If pieces do not fit, check that whole sides touch.

36:00 – 40:00 Brain break

script 3:10 of 4:00

36:00	**SLIDE**	Show slide **Shape moves** (s-break)
36:08	**REC**	*Recording: stand up and join in! Make the shapes with your arms and body as I call them.*
36:17	**SAY**	Stand up and get a sip of water. I call a shape and you make it with your arms or your body. A triangle: hands together above your head. A rectangle: stand tall, then stretch your arms out wide. A circle: a big ring with your arms.
36:40	**CHILDREN**	Call triangle, rectangle, circle and square one at a time, then call 2 shapes at once, one with each arm. Children make each shape; a child who sits makes the shapes with their hands. [2:30]

40:00 – 70:00 Workbook

script 21:12 of 30:00

40:00	**SLIDE**	Show slide **Workbook page 1: Start here** (s-ws1)
40:08	**REC**	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*
40:17	**READ**	Page 1, Start here. Question 1: look at the picture of 8 shapes. Count each kind and fill in the blanks. Question 2: a square is cut into 2 triangles. Draw the line with your ruler. Then circle half or quarter.
40:39	**CHILDREN**	Page 1, questions 1 and 2. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [6:40]
47:19	**SLIDE**	Show slide **Check page 1** (s-ws1ans)
47:27	**READ**	Question 1: 4 squares, 3 triangles, 1 circle, and 1 more square than triangles. Question 2: a line from one corner to the opposite corner, and each triangle is a half. Tick your own page with a tick.
47:48	**SAY**	After question 2 you may choose: carry on, or try Think harder. Everyone may choose, at any time.
47:57	**SLIDE**	Show slide **Workbook page 2: Start here** (s-ws2)

48:05	READ	Page 2. Question 3: build the boat with squares and triangles. Use as few triangles as you can. Fill in how many of each. Question 4: the rectangle is made of squares. Fill in the squares in each row, the rows, and the squares in all. Question 5: fill in what 2 squares make, what 2 half circles make and what 2 quarter circles make.
48:41	CHILDREN	Page 2, questions 3 to 5. [7:00]
55:41	SLIDE	Show slide **Check page 2** (s-ws2ans)
55:49	READ	Question 3: 2 triangles and 2 squares. Question 4: 4 squares in each row, 3 rows, 12 squares in all. Question 5: a rectangle; a circle; a half circle. Tick your own page with a tick.
56:08	SLIDE	Show slide **Choose your next page** (s-choice)
56:16	SAY	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time. Anything not finished is for home too.
56:32	MOVE	Stand and reach up for 30 seconds, then sit down and choose.
57:02	CHILDREN	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. If Start here took the whole time, say that pages 3 to 5 are for home, and keep the time. [4:10]
	IF STUCK	Q1: count one kind at a time and touch each shape. Q2: a line from one corner to the opposite corner makes 2 triangles. Q3: the slanted ends of the boat need triangles. Q4: count the squares in one row, then count the rows. Q5: try the pieces with whole sides touching.
	IF STUCK	A child who is still stuck ticks I am still stuck on the look-back page. The grown-up can photograph that page after class and send it to the business WhatsApp (never a personal phone). You then help in the open session in the next lesson. Never a private message to a child.
	NOTE	Think harder splits a figure with a ruler, counts shapes of each kind into a table and has the beyond puzzle Every Triangle. Go further has the fewest pieces, which sets of pieces make a figure, and a figure to draw for a partner (beyond Primary 1 on purpose). Do not announce who is on which page. Start here takes a middle child about 16 minutes, so the choice time is short: for most children pages 3 to 5 are for home. Say so plainly, do not skip the answer slides, and take any extra minutes from the flex time.

70:00 – 76:00 Talk time

script 4:22 of 6:00

70:00	SLIDE	Show slide **Talk and explain** (s-talk)
70:25	REC	*Recording: pause. Tell a grown-up whether 2 triangles always make a square, and test it with your pieces.*
70:34	READ	A child says: 2 triangles always make a square. Is the child right? Test it with your triangles.
70:44	CHILDREN	Test it with your 2 triangles: try touching different sides. [0:40]
71:24	CHILDREN	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [0:40]
72:04	ASK	Who would like to share? Volunteers first. A pass is always fine. [wait about 80 s for answers]
73:30	SLIDE	Show slide **Back to the sneak-peek** (s-sneakans)
73:55	SAY	Remember the sneak-peek? Puzzle one: 2 triangles. Puzzle two: many answers are right. It depends which sides touch.
74:04	ASK	Did anyone change their mind while we worked? You can just nod. [wait about 12 s for answers]
	NOTE	Model answer: no. 2 triangles can make a square when the longest sides touch, a bigger triangle when 2 short sides touch, and a slanted four-sided shape. It depends on which sides touch. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed.

76:00 – 82:00 Exit ticket

script 4:34 of 6:00

76:00	SLIDE	Show slide **Exit ticket** (s-exit)

76:25	**REC**	*Recording: pause and do the exit ticket on your own. Then press play to check it.*
76:33	**SAY**	Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5.
76:41	**READ**	E1. Mei makes a picture with 3 squares, 2 triangles and 2 circles. How many shapes are in the picture? Fill in the blank. E2. Can 2 triangles make a square? Circle Yes or No. E3. A rectangle is made of 6 squares. Mei makes it again with rectangles that are each 2 squares long. How many rectangles does she need? Fill in the blank.
77:16	**CHILDREN**	Children answer E1 to E3 on their own. [2:45]
80:01	**SLIDE**	Show slide **Check the exit ticket** (s-exitans)
80:26	**READ**	E1: 7. E2: Yes. E3: 3.
80:30	**SAY**	Tick your own ticket with a tick. Well done.
	NOTE	Nothing is collected or marked live. Children tick their own ticket. If many children miss one question, spend two minutes on it at the start of the next lesson, in the open session, no names. Do not read out any child result.

82:00 – 85:00 Look-back

script 2:23 of 3:00

82:00	**SLIDE**	Show slide **Look-back** (s-look)
82:25	**REC**	*Recording: pause and fill in your look-back box. You can pass.*
82:30	**SAY**	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out. If you tick I am still stuck, ask your grown-up to take a photo of this page after class and send it to our business WhatsApp. Then we will help you.
83:03	**CHILDREN**	Children fill in the look-back box on page 5. [1:20]

85:00 – 90:00 Flex time and goodbye

script 0:30 of 5:00

85:00	**SLIDE**	Show slide **Well done, inquirers!** (s-end)
85:08	**SAY**	Well done, inquirers. Next lesson is Copy on the Grid. Nothing needs cutting. Please have the grid page ready, a button for a counter, a ruler and a pencil. Keep your shape pieces in their envelope. Goodbye, and thank you to the grown-ups.
	SLIDE	Show slide **Optional: 4 squares, one rectangle** (s-opt) *(optional)*
	SAY	Optional, only if you are on time: Mei has 4 squares. She makes a bigger square with all 4. Can she make it with fewer pieces?
	CHILDREN	Short think time, then take two answers. Children try with their pieces. [1:30]
	SLIDE	Show slide **If we have time: pick one** (s-menu) *(optional)*
	SAY	Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).
	NOTE	This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: 4 squares make a bigger square, and 2 rectangles can make it with 2 pieces instead of 4. If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.

3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content. If you are more than 3 minutes ahead at minute 40, use one menu activity then.
- A few children ahead: a few children may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask "How do you know?" and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

Activity	Minutes	You say	Recording cue	Answer
Is it always true?	4 (optional)	A child says: 2 squares always make a rectangle. Is that always true? Try putting the 2 squares together in different ways.	*Recording: pause. Do 2 squares always make a rectangle? Try putting them together in different ways.*	Not always. 2 squares make a rectangle when whole sides touch. If only a corner touches, or one square is slid along the other, the shape is not a rectangle.
Make a puzzle for your grown-up	4 (optional)	Build a figure with 5 pieces. Ask a grown-up to count each kind of shape and say how many pieces in all. Check together.	*Recording: pause. Build a figure with 5 pieces. Ask a grown-up to count each kind of shape.*	No single answer. A good puzzle has pieces with whole sides touching, and the count of each kind adds up to 5.
Teach it back	3 (optional)	Pretend your grown-up or a toy has never put shapes together. Show how 2 triangles make a square: the 2 longest sides touch.	*Recording: pause. Show a grown-up how 2 triangles make a square, with the 2 longest sides touching.*	Put the 2 longest sides of the 2 triangles together. The 2 triangles make a square.

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 8 to 12)

Puzzle 1 (access): Look at the picture. How many triangles are there? Write the number.

> **Answer: 2**

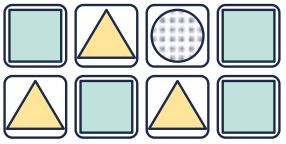
Puzzle 2 (think): Put 2 triangles from your kit together so that whole sides touch. Name one shape you can make. Draw it.

> **Answer: A square, or a bigger triangle, or a slanted four-sided shape. Many answers are right: it depends on which sides touch.**

Unmarked. Never put answers on boards in front of the group and never read a name with an answer. Reveal the answers where your script says (usually Talk time).

Start here (workbook pages 1 and 2)

S1 **Look at the picture of 8 shapes. Count each kind. Fill in the blanks.**

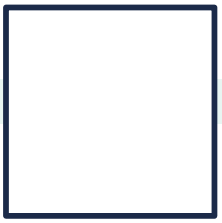


(a) How many squares? ☐
(b) How many triangles? ☐
(c) How many circles? ☐
(d) How many more squares than triangles? ☐

> **Answer: 4 squares; 3 triangles; 1 circle; 1 more square than triangles**

Hint (say this first): Count one kind at a time. Touch each shape as you count it, so you do not count one twice.
Worked solution: There are 4 squares, 3 triangles and 1 circle: 4 + 3 + 1 = 8 shapes. 4 squares is 1 more than 3 triangles.
Common mistakes: Counting a shape twice or missing one.; Writing the total, 8, in one of the blanks.; Writing 4 for the difference.
Time for a middle child: about 3.5 min.

S2 **A square is cut into 2 triangles. Use your ruler. Draw the line.**



(a) Draw 1 line on the square to make 2 triangles.
(b) Each triangle is a half or a quarter of the square. Circle one.

> **Answer: a line from one corner to the opposite corner; half**

Hint (say this first): The 2 triangles must be the same size. Draw the line from one corner to the opposite corner.
Worked solution: A line from one corner to the opposite corner cuts the square into 2 triangles that match. Each triangle is 1 of 2 equal parts, so each is a half of the square.
Common mistakes: Drawing a line across the middle: that makes 2 rectangles.; Writing quarter: there are only 2 parts.
Time for a middle child: about 3 min.

S3 **Build the boat with squares and triangles. Use as few triangles as you can. Fill in the blanks.**

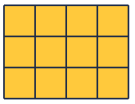


(a) The boat uses ☐ triangles.
(b) The boat uses ☐ squares.

> **Answer: 2 triangles; 2 squares**

Hint (say this first): Look at the 2 slanted ends of the boat. Which piece has a slanted side? Then fill the middle with squares.
Worked solution: Each slanted end needs 1 triangle, so there are 2 triangles. The middle is 2 squares side by side. The boat is 2 triangles and 2 squares.
Common mistakes: Using squares for the slanted ends.; Writing 4 triangles (2 more for the middle).
Time for a middle child: about 4 min.

S4 **The rectangle is made of squares. Fill in the blanks.**



(a) There are ☐ squares in each row.
(b) There are ☐ rows.
(c) There are ☐ squares in all.

Answer: 4; 3; 12

Hint (say this first): Count the squares in one row first. Then count the rows. Then count all the squares.
Worked solution: There are 4 squares in each row and 3 rows: 4, 8, 12. So there are 12 squares in all.
Common mistakes: Writing 7 (4 + 3).; Counting only one row.
Time for a middle child: about 3 min.

S5 **Fill in the blanks. Circle one word each time.**

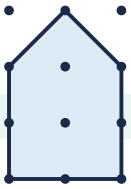
(a) 2 squares make a ____ . Circle one.
(b) 2 half circles make a ____ . Circle one.
(c) 2 quarter circles make a ____ . Circle one.

Answer: rectangle; circle; half circle

Hint (say this first): Try it with your pieces. Put the 2 pieces together so that whole sides touch.
Worked solution: 2 squares side by side make a rectangle. 2 half circles with their straight sides together make a circle. 2 quarter circles with their straight sides together make a half circle.
Common mistakes: Writing square for 2 squares.; Writing circle for 2 quarter circles: it takes 4.
Time for a middle child: about 2.5 min.

Think harder (page 3): the taught idea in a new setting

T1 **Use your ruler. Split the figure into its shapes.**



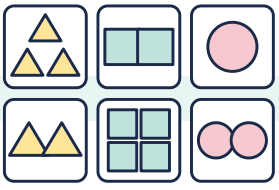
(a) Draw 1 line to split the figure into 1 triangle and 1 square.
(b) Draw 2 more lines to split the square into 4 small squares.

Answer: a line across at the bottom of the roof; 2 more lines through the middle

Hint (say this first): The roof is the pointed shape on top. Where does the roof end and the square begin? Then think about 4 small squares inside the big square.
Worked solution: Draw a line across from one side of the house to the other, where the roof ends. That splits the figure into 1 triangle and 1 big square. Then draw 1 line down the middle and 1 line across the middle of the big square: that makes 4 small squares.
Common mistakes: Drawing a line down the middle of the whole house.; Drawing only 1 extra line in the square.
Time for a middle child: about 5 min.

T2 **Count each kind of shape in the picture. Fill in the table.**

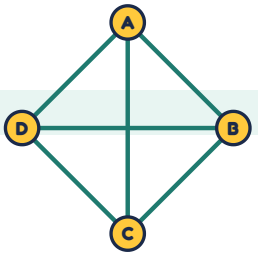
Triangles	Squares	Circles
☐	☐	☐



Answer: triangles 5; squares 6; circles 3

Hint (say this first): Count one kind at a time. Each tile has 1 to 4 shapes of the same kind. Add up the shapes, not the tiles.
Worked solution: Triangles: 3 and 2 make 5. Squares: 2 and 4 make 6. Circles: 1 and 2 make 3. 5 + 6 + 3 = 14 shapes in all.
Common mistakes: Counting the tiles instead of the shapes.; Counting a tile that has 4 shapes as 1.
Time for a middle child: about 5 min.

T3 **Every Triangle. The square is turned on its corner. It has 2 lines from corner to corner. Count the small triangles and the big triangles. Fill in the blank.**



There are ☐ triangles in all.

Answer: 8

Hint (say this first): First count the small triangles next to the middle. Then count the big triangles that use 3 corners of the square.

Worked solution: There are 4 small triangles, each with the middle point as one corner. There are 4 big triangles, each made of 2 small ones and using 3 corners of the square: A B C, B C D, C D A and D A B. 4 and 4 make 8 triangles in all.

Common mistakes: Counting only the 4 small triangles.; Counting a big triangle twice.

Time for a middle child: about 6 min.

Go further (page 4): mastery and extension

G1 **The rectangle is 2 rows of 3 squares. Build it with the fewest pieces. Use squares and rectangles from your kit.**



(a) With squares only it takes ☐ pieces.

(b) With the fewest pieces it takes ☐ pieces.

Answer: 6; 3

Hint (say this first): A rectangle in the kit is as long as 2 squares. How many of them cover 6 squares?

Worked solution: With squares only it takes 6 pieces. A kit rectangle covers 2 squares, so 3 rectangles cover 6 squares. 3 pieces is the fewest, because no piece covers more than 2 squares.

Common mistakes: Writing 6 for the fewest.; Writing 2 (2 rectangles cover only 4 squares).

Time for a middle child: about 5 min.

True/False statements: 6 squares.; 3 rectangles.; 2 rectangles and 2 squares.; 2 rectangles and 3 squares.

G2 **The staircase is made of 6 squares. Which sets of pieces make it? Tick True or False for each set.**



Answer: true; false; true; false

Hint (say this first): Try each set with your pieces. A rectangle covers 2 squares that touch side by side. Check that no square of the staircase is left out.

Worked solution: 6 squares make it: True. 3 rectangles cannot make it: after the top square and the bottom right square each take a neighbour, the 2 squares left over do not touch, so they need 2 squares, not 1 rectangle: False. 2 rectangles and 2 squares cover 4 + 2 = 6 squares: True. 2 rectangles and 3 squares cover 7 squares, too many: False.

Common mistakes: Saying True for any set with the right number of squares but the wrong shape.; Not checking that the total is 6.

Time for a middle child: about 6 min.

G3 **Draw a figure for a partner. It has exactly 2 triangles, 1 square and 1 rectangle. Many answers are right.**

Answer: any figure with exactly 2 triangles, 1 square and 1 rectangle

Hint (say this first): Draw the pieces touching with whole sides. Then count each kind. Check: 2 triangles, 1 square, 1 rectangle.

Worked solution: A house works: a rectangle for the wall, a square for a window, and 2 triangles for the roof. Count each kind to check your figure: 2 triangles, 1 square, 1 rectangle.

Common mistakes: Drawing 2 rectangles.; Not counting the shapes at the end.

Time for a middle child: about 5 min.

Investigation (page 5): 2 triangles, many shapes

Easy first step: Put 2 triangles together so whole sides touch. Make a square. Draw it.
Then: Find other shapes you can make with 2 triangles. Draw each one. How many different shapes can you find?

> **Model answer: First: a square (the 2 longest sides touch). Then 3 different shapes in all when whole sides touch: a square, a bigger triangle (2 short sides touch, the right-angle corners together) and a slanted four-sided shape (2 short sides touch, turned the other way). A child who finds 2 has the main idea; a child who finds the third has seen that it depends on which sides touch.**

Look for: A child who tries different sides and draws each result, who says the shape depends on which sides touch, and who checks that whole sides touch. Credit any clear reason. Do not mark a child down for stopping at 2 shapes.
Common mistakes: Letting only a corner touch.; Drawing the same shape twice (a turned square is the same square).

Talk and explain (minutes 70 to 76)

A child says: 2 triangles always make a square. Is the child right? Test it with your triangles. Sentence starter: I think ... because ...

> **Model answer: No. 2 triangles can make a square when the 2 longest sides touch, a bigger triangle when 2 short sides touch, and a slanted four-sided shape. It depends on which sides touch.**

Look for: A child who tests it with the pieces and finds a shape that is not a square, or says it depends on which sides touch. Not just yes or no. Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 76 to 82): 3 questions

E1 **Mei makes a picture with 3 squares, 2 triangles and 2 circles. How many shapes are in the picture? Fill in the blank.**

There are ☐ shapes in the picture.

> **Answer: 7**

Worked solution: 3 squares, 2 triangles and 2 circles: 3 + 2 + 2 = 7 shapes.
Common mistakes: Writing 3, the squares only.; Writing 5 (leaving out the circles).

E2 **Can 2 triangles make a square? Circle Yes or No.**

> **Answer: Yes**

Worked solution: Yes: when the 2 longest sides touch, 2 triangles make a square.
Common mistakes: Saying No because they are not squares.; Not trying it with the pieces.

E3 **A rectangle is made of 6 squares. Mei makes it again with rectangles that are each 2 squares long. How many rectangles does she need?**

She needs ☐ rectangles.

> **Answer: 3**

Worked solution: Each rectangle covers 2 squares. 2, 4, 6: she needs 3 rectangles.
Common mistakes: Writing 6, the squares.; Writing 2.

Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (putting pieces together; counting each kind; splitting a figure; something else); *What helped me* (my shape pieces; touching each shape as I count; checking my answer; I was not stuck); *Next time I will try* (trying the pieces first; counting one kind at a time; checking my answer; asking a question); *I am still stuck (my grown-up will photograph this page for my tutor)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer
H1	Start here	A picture has 5 circles and 2 squares. There are ____ more circles than squares.	3
H2	Start here	2 squares make a ____ . Write rectangle, circle or triangle.	rectangle
H3	Think harder	A square is cut corner to corner into 2 triangles. 4 squares are cut this way. How many triangles are there? ____	8
H4	Go further	A rectangle is made of 2 rows of 4 squares. Rectangles that are 2 squares long cover it. The fewest pieces is ____ .	4
H5	Start here	Ali has 14 pencils. Mei has 9 pencils. Draw a strip for each. Ali has ____ more pencils than Mei.	5 more pencils

Online practice (`out/l32/practice`, key `t4l32`): 13 items; 4 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

6. Parent line for the term report

Your child is learning to build figures from shapes and to count the shapes inside a figure. This week: squares, triangles and rectangles in pictures.

Term line (plan, section 5): Your child is learning to choose a way to solve a problem, and to make problems of their own. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–2	**Build the Figure** s-title	TODAY WE WILL: / Put shapes together. / Find the shapes inside a figure.	Lesson 2 of Term 4, 90 minutes. Objective: put 2 triangles together to make a square or a bigger triangle, build a figure from squares and triangles, count each kind of shape in a figure, and meet the beyond-Primary-1 puzzle Every Triangle. Plain statement for families: counting and naming shapes in a picture is Primary 1 work, and so is putting shapes together; counting every triangle of every size, finding the fewest pieces and listing which sets of pieces make a figure go past the school Primary 1 syllabus on purpose, at Think harder and Go further only. Time plan: opening 0 to 2, warm-up 2 to 8, sneak-peek 8 to 12, explore 12 to 24, mini-lesson 24 to 36, brain break 36 to 40, workbook 40 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Say the recording notice in one line before you start the recording: the tutor, the shared screen and any voice that is not muted are recorded; a child may keep the camera off. Think harder, Go further and the Investigation are for every child to choose.
2	2–8	**Which shape am I?** s-rem1	[picture: shapes2d] / [picture: shapes2d] / [picture: shapes2d] / Which shape has 3 corners? Which has 4 sides that are all the same? Which has 2 long sides and 2 short sides?	Three shapes labelled A, B and C: A is a triangle, B a square, C a rectangle. Ask each clue and let children answer on three or with fingers (1, 2 or 3 for A, B, C): 3 corners is A, the triangle; 4 equal sides is B, the square; 2 long sides and 2 short sides is C, the rectangle. Then ask for the names. The slide never shows the names. This is the holiday-course naming, taught again with clues so a child who joins cold can follow. A child who is unsure touches the sides with a finger and counts the corners.
3	2–8	**Fold a square corner to corner** s-rem2	Take a paper square. Fold it corner to corner. / Open it. What 2 shapes do you see?	Children fold a paper square (a spare one from last lesson, or any paper square) corner to corner, open it and trace the fold. The fold cuts the square into 2 triangles, and each triangle is a half of the square. Ask: how many shapes, what are they called, are they the same size (lay one on the other). This is the first lesson of Term 4 taught again. A child with no paper square draws a square and a line from one corner to the opposite corner. The slide never shows the answer. Answer: 2 triangles.
4	2–8	**Count by rows** s-rem3	[picture: grid] / How many squares make the rectangle? Count by rows.	A rectangle made of 2 rows of 3 squares. Children count the squares row by row: 3, then 3 more, 6 in all. This is the array idea from Term 2 Lesson 5, taught again so that counting squares by rows is easy later. Ask: how many in each row, how many rows, how many in all. The slide never shows the count. A child who is unsure points to each square and counts in ones. Answer: 6.
5	8–12	**Sneak-peek puzzles** s-sneak	PUZZLE 1: anyone can try: / Look at the picture. How many triangles are there? / [picture: panel] / PUZZLE 2: think: / Put 2 triangles from your kit together so that whole sides touch. Name one shape you can make. Draw it.	Puzzle 1 is an access question: any child can count the triangles in a row of 5 pictures (the 2 triangles are different colours and one has stripes, so the count does not depend on colour). Puzzle 2 is a thinking question on the lesson idea: 2 triangles put together make a square, a bigger triangle or a slanted four-sided shape, depending on which sides touch. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write and draw on their own paper. Nobody holds paper to the camera. There is no private chat. Answers: 2; many answers are right. Do not reveal puzzle 2 until Talk time. Do not write scores or keep a list.

6	12–24	**Explore: 2 triangles** s-explore1	Take 2 triangles from your kit. / Put them together so whole sides touch. / Make a square. / Now make a bigger triangle. / Whole sides must touch.	Round 1, about 4 minutes. Each child has 2 triangles from the kit page. Ask them to put the 2 triangles together so that a whole side touches a whole side (not a corner and not half a side). They find a square (the 2 longest sides touch) and a bigger triangle (2 short sides touch, with the right-angle corners on the same side). Some find a slanted four-sided shape: praise it, it is a third way. Ask: what shape did you make; which sides touched. The answer pictures are on the next slide, after the children have answered. Answers: a square; a bigger triangle. A child with no triangles cuts a paper square corner to corner (the grown-up cuts). Nobody holds a figure up to the camera: children say what they made.
7	12–24	**Explore: 2 triangles** s-explore1-ans	Take 2 triangles from your kit. / Put them together so whole sides touch. / Make a square. / Now make a bigger triangle. / Whole sides must touch. / [picture: shapes2d] / [picture: grid] / 2 TRIANGLES MAKE: / a square. / a bigger triangle.	Answer slide for the previous slide. Read the answer aloud after the children have answered, then give the idea in your own words. The notes on the previous slide have the answers and the common slips.
8	12–24	**Explore: build the house** s-explore2	Look at the house. / Build it with your squares and triangles. / Which pieces did you use? / [picture: grid]	Round 2, about 5 minutes. The house is drawn on dots; each square on the dot grid is one square piece and each slanted half-square is one triangle. Children build the house on the table with their pieces, check it matches the picture, and say which pieces they used: 2 squares for the bottom and 2 triangles for the roof, 4 pieces in all. Ask: which pieces did you use; how many of each; how did you know the roof needs triangles (the slanted sides). The answer is on the next slide. Some children use 1 rectangle for the bottom: that works and is a good idea, so ask whether it uses more or fewer pieces; the fewest-pieces question comes back at Go further. Answers: 2 squares and 2 triangles, 4 pieces.
9	12–24	**Explore: build the house** s-explore2-ans	Look at the house. / Build it with your squares and triangles. / Which pieces did you use? / [picture: grid] / THE HOUSE USES: / 2 squares and 2 triangles. / That is 4 pieces.	Answer slide for the previous slide. Read the answer aloud after the children have answered, then give the idea in your own words. The notes on the previous slide have the answers and the common slips.
10	12–24	**Explore: your own picture** s-explore3	Make a picture with your pieces: a robot, a rocket or a boat. / Use at least 3 different kinds of shape. / Count how many of each shape you used. / Say it: My picture has … squares and … triangles. / Keep it on the table. Say it. Nobody holds it up.	Round 3, about 3 minutes. Children make any picture they like with up to 8 pieces. They count each kind and say it aloud to a grown-up or to the screen: my picture has 2 squares, 3 triangles and a circle. Volunteers say their count; nobody holds a figure up to the camera and nobody is compared. Ask: which kind did you use most; how many pieces in all. A child who needs a start builds a robot: a rectangle body, a square head and 2 triangle feet. Praise any picture; the point is counting by kind. No answers on the slide.
11	24–36	**2 shapes make 1 shape** s-m1	2 SQUARES: / make a rectangle. / [picture: grid] / 2 HALF CIRCLES: / make a circle. / [picture: shapes2d] / 2 TRIANGLES: / make a square or a bigger triangle. / [picture: shapes2d]	Three ways that 2 pieces make 1 shape, with whole sides touching. 2 squares side by side make a rectangle: show it with 2 kit squares. 2 half circles with their straight sides together make a circle: show it with the 2 half circles. 2 triangles make a square or a bigger triangle, as in Explore. Children try each with their pieces. Ask: what do 2 squares make (a rectangle); what do 2 half circles make (a circle); what do 2 triangles make (it depends on which sides touch). Say: the shapes must touch with whole sides. The pictures show the made shape.

12	24–36	**Find the shapes inside** s-m2	Look at the house. Count each kind of shape. / [picture: shapes2d] / How many triangles? / How many rectangles? / How many squares?	Now we look inside a figure. The house is made of shapes: the roof is a triangle, the wall and the door are rectangles, the window is a square. Ask each question, then wait: how many triangles (1), how many rectangles (2), how many squares (1). Children point to each shape on their own screen and count aloud on three. The answer appears on the next slide, after the children have answered. A child who says 2 squares is asked whether the window is as tall as it is wide. Answers: 1 triangle; 2 rectangles; 1 square. Do not name a child with a wrong answer.
13	24–36	**Find the shapes inside** s-m2-ans	Look at the house. Count each kind of shape. / [picture: shapes2d] / How many triangles? / How many rectangles? / How many squares? / 1 triangle, 2 rectangles and 1 square. The roof is a triangle. The wall and the door are rectangles. The window is a square.	Answer slide for the previous slide. Read the answer aloud after the children have answered, then give the idea in your own words. The notes on the previous slide have the answers and the common slips.
14	24–36	**Count each kind** s-m3	Count each kind. Which kind is there most of? / [picture: panel] / How many triangles? How many circles? How many squares?	A row of shapes to count by kind. Count the triangles one at a time, touching each (children point to their screen), then the circles, then the squares. Say a method: count one kind at a time, and make a mark so that you never count one twice. Ask: how many triangles, how many circles, how many squares, how many shapes in all, which kind is most. The answer appears on the next slide. Answers: 4 triangles, 1 circle, 1 square: 6 in all; the triangles are the most.
15	24–36	**Count each kind** s-m3-ans	Count each kind. Which kind is there most of? / [picture: panel] / How many triangles? How many circles? How many squares? / 4 triangles, 1 circle and 1 square. There are 6 shapes in all. There are most triangles.	Answer slide for the previous slide. Read the answer aloud after the children have answered, then give the idea in your own words. The notes on the previous slide have the answers and the common slips.
16	24–36	**A rectangle can replace 2 squares** s-m4	4 SQUARES: / make a long rectangle. / That is 4 pieces. / [picture: grid] / 2 RECTANGLES: / make the same long rectangle. / That is 2 pieces. / A rectangle is as long as 2 squares.	The same figure can be made with different pieces. Show 4 squares in a row, and then the same row made from 2 rectangles of your kit, each 2 squares long. Ask: how many pieces with squares (4); how many pieces with rectangles (2); which uses fewer pieces (the rectangles). Say: a rectangle in our kit is as long as 2 squares. This is the idea of the fewest pieces, which comes back at Go further. Children lay their pieces on the table to check. Answers: 4 pieces; 2 pieces.
17	24–36	**Your turn** s-m5	ITEM 1: / A picture has 3 circles and 2 squares. How many shapes are there? / ITEM 2: / Can 2 squares make a triangle? Say Yes or No.	Two quick items, answered on the children's own paper or said quietly on three (say so aloud). Item 1: 3 circles and 2 squares make 5 shapes in all. Item 2: 2 squares cannot make a triangle: put 2 squares side by side and you get a rectangle. Walk the cameras and wait 10 seconds before helping. Nobody holds paper up. The next slide shows the answers and children tick their own work. Answers: 5; No.
18	24–36	**Your turn** s-m5-ans	ITEM 1: / A picture has 3 circles and 2 squares. How many shapes are there? / ITEM 2: / Can 2 squares make a triangle? Say Yes or No. / Item 1: 5 shapes. Item 2: No. 2 squares make a rectangle, not a triangle.	Answer slide for the previous slide. Read the answer aloud after the children have answered, then give the idea in your own words. The notes on the previous slide have the answers and the common slips.

19	36–40	**Shape moves** s-break	Stand up. Make each shape I call. / Triangle. Rectangle. Circle. / Sip some water.	Four minutes. Stand up with a little clear space. Call the shapes one at a time and children make them with their arms and body: a triangle (hands together above the head like a roof), a rectangle (stand tall and then stretch the arms out wide), a circle (arms in a big ring in front), a square (elbows out). Then call 2 shapes at once: make a triangle with one arm and a circle with the other. A child may stay seated and make the shapes with their hands. No equipment, no running. Do not give prizes or compare children.
20	40–70	**Workbook page 1: Start here** s-ws1	Questions 1 and 2. Take your time. / questions S1, S2	Read questions 1 and 2 aloud. Children work on their own with their pieces; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 2 any child may choose to carry on or try Think harder. Question 1: count each kind of shape in a picture and say how many more of one kind. Question 2: draw a line on a square to show 2 triangles, and say what part of the square each triangle is.
21	40–70	**Check page 1** s-ws1ans	answers S1, S2	Read each answer aloud. Children tick their own page with a tick. Never ask a child to show a wrong answer to the group. Answers: Q1 4 squares, 3 triangles, 1 circle, 1 more square than triangles; Q2 a line from one corner to the opposite corner, and each triangle is a half of the square.
22	40–70	**Workbook page 2: Start here** s-ws2	Questions 3 to 5. Take your time. / questions S3, S4, S5	Read questions 3 to 5 aloud, one at a time. Question 3: build the boat with squares and triangles, using as few triangles as you can, and say how many of each. Question 4: a rectangle made of squares in rows: how many in each row, how many rows, how many in all. Question 5: what do 2 squares, 2 half circles and 2 quarter circles make. Children may use their pieces. Wait 10 seconds before helping.
23	40–70	**Check page 2** s-ws2ans	answers S3, S4, S5	Read each answer aloud so a child watching the recording can tick their own work. Never ask a child to show a wrong answer to the group. Answers: Q3 2 triangles and 2 squares; Q4 4 squares in each row, 3 rows, 12 squares in all; Q5 2 squares make a rectangle, 2 half circles make a circle, 2 quarter circles make a half circle.
24	40–70	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | Split a figure. Every Triangle / Go further (page 4) | The fewest pieces / Investigation (page 5) | 2 triangles, many shapes / Check | Finish and check your own work / Everyone may choose. Not finished? Pages 3 to 5 are for home too.	Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 5 minutes.
25	70–76	**Talk and explain** s-talk	A CHILD SAYS: / 2 triangles always make a square. / Is the child right? Test it with your triangles. / I think … because …	One minute of talk to the grown-up next to you, or to your screen. Children test it with their 2 kit triangles: try touching different sides. Then volunteers first; a pass is always fine; never call a child by name to answer. Model answer: no. 2 triangles can make a square when the 2 longest sides touch, a bigger triangle when 2 short sides touch, and a slanted four-sided shape. It depends on which sides touch. Repeat each child answer in your own words for the recording. The child who is wrong is a child in another class: no cast name is given the wrong answer.
26	70–76	**Back to the sneak-peek** s-sneakans	Puzzle 1: 2 triangles. Puzzle 2: a square, a bigger triangle or a slanted four-sided shape. Many answers are right.	Reveal both puzzles now. Say: many of you found a good shape. Do not read out anyone earlier answer. Puzzle 1: the row has 2 triangles; the squares and the circle are not triangles. Puzzle 2: 2 triangles make a square when the longest sides touch, and a bigger triangle when 2 short sides touch; some children found a slanted four-sided shape. It was the same idea as the child in the talk question.

27	76–82	**Exit ticket** s-exit	On your own. No looking back. / questions E1, E2, E3	Children answer on page 5. Read E1, E2 and E3 aloud, and the words Yes and No. Children tick their own ticket against the next slide. Nobody holds paper to the camera and there is no private chat.
28	76–82	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children tick their own ticket with a tick. Answers: 7; Yes; 3. Do not read out any child result.
29	82–85	**Look-back** s-look	[look-back box]	Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, a grown-up may photograph that page after class and send it to the business WhatsApp (the number is in the parent summary, never on a slide). You then help in the open session next lesson. Never a direct message to a child. The Next time I will try row is optional.
30	85–90	**Well done, inquirers!** s-end	Show a grown-up one figure you built today. / NEXT LESSON: / Copy on the Grid / Have ready: the grid page (it is not cut), a button for a counter, a ruler and a pencil.	Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (the look-back). Do not start anything new. Remind families that the next lesson needs nothing cut: the grid page in the Term 4 book is for drawing on, a button or a small coin is the counter, and a ruler and a pencil. Keep the shape pieces in their envelope. If the whole class is ahead, show the optional slide and use the Spare-time menu from the tutor guide.
31	optional	**Optional: 4 squares, one rectangle** s-opt	Mei has 4 squares. She makes a square with all 4. / What shape is it? Can she make it with fewer pieces?	Optional, only if you are on time. 4 squares in a 2 by 2 block make a big square. With the kit rectangles (each 2 squares long) she can make it with 2 rectangles: 2 pieces instead of 4. Children try with their pieces. Take two answers and move on. This links to the fewest pieces at Go further. Answers: a bigger square; yes, 2 rectangles. There is no picture on purpose: children build it.
32	optional	**If we have time: pick one** s-menu	Activity | What we do / Is it always true? | Do 2 squares always make a rectangle? / Make a puzzle | Build a figure. Ask: how many of each shape? / Teach it back | Show a toy how 2 triangles make a square. / Optional. Everyone may join.	Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get a pencil, a ruler, paper and your 20 shape pieces. Pause or rewind any time.*	Gather your shape pieces, pencil, ruler and paper. Press play when ready.
2:00	Remember warm-up	*Recording: join in! Name each shape, fold a paper square corner to corner, and count the squares by rows.*	Play along. Fold a paper square corner to corner. Say each answer out loud.
8:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles. Use 2 triangles for the second one. Guessing is fine. Press play for the next part.*	Pause. Try both puzzles with your pieces. A guess is fine.
12:00	Explore	*Recording: pause now. Put 2 triangles together, build the house from the picture, then make your own picture.*	Pause. Make a square from 2 triangles. Build the house. Make your own picture and count each shape.

24:00	Mini-lesson	*Recording: keep watching and copy the examples. Rewind if you miss a step.*	Watch and copy the examples with your pieces. Rewind any step you missed.
36:00	Brain break	*Recording: stand up and join in! Make the shapes with your arms and body as I call them.*	Stand up. Make each shape with your arms as it is called.
40:00	Workbook	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*	Pause. Do page 1, then play to check. Then page 2. Then choose a page to try.
70:00	Talk time	*Recording: pause. Tell a grown-up whether 2 triangles always make a square, and test it with your pieces.*	Pause. Explain your answer to someone at home and test it with your pieces.
76:00	Exit ticket	*Recording: pause and do the exit ticket on your own. Then press play to check it.*	Pause. Do the exit ticket alone, then play to check it.
82:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.
85:00	Flex time and goodbye	*Recording: well done for finishing! Show a grown-up one figure you built.*	Show a grown-up your workbook and one figure you built.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	2:00	1:41	1:54	ok
remember	6:00	4:58	5:54	ok
sneak	4:00	2:57	3:27	ok
explore	12:00	8:38	10:15	ok
mini	12:00	9:24	11:06	ok
break	4:00	3:10	3:45	ok
work	30:00	21:12	25:13	ok
talk	6:00	4:22	5:11	ok
exit	6:00	4:34	5:23	ok
look	3:00	2:23	2:48	ok
flex	5:00	0:30	0:33	ok
Total	90:00	63:48	75:30	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 5 questions; a middle child needs about 6.5 min for page 1 and 9.5 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.